

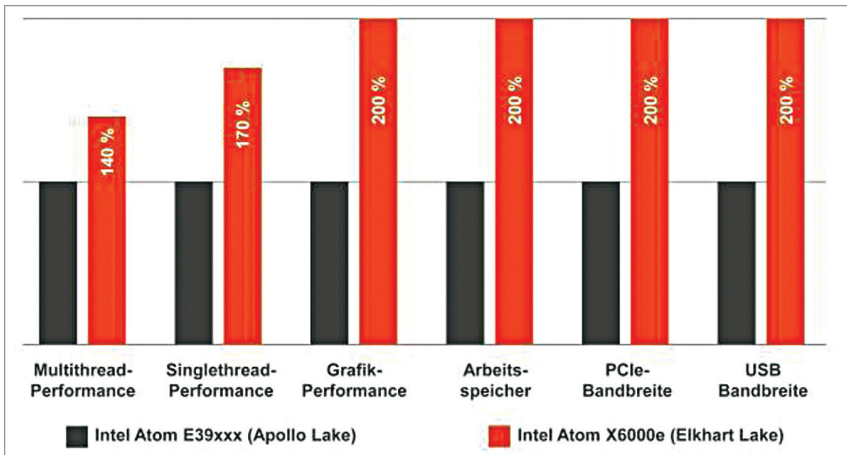


Fig. 1: Drastic improvements in per watt performance, compared to Apollo Lake processors, are provided by congatec boards and modules featuring Intel Elkhart Lake processors

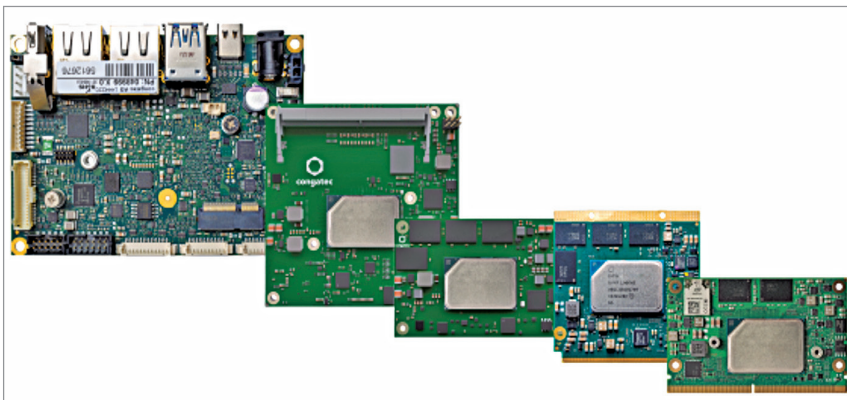


Fig. 2: Developers of small form factor designs that feature advanced graphics and computing performance, together with a wide range of I/O with a low-power envelope, will benefit from congatec's Pico-ITX to COM-Express Compact and Mini, SMARC, and Qseven modules that feature the new Intel Atom processors

pared to only 18 EUs found in the low-power Apollo Lake Gen9 unit. Moreover, improved compression, increased L3 cache, and higher data transfer rates lead to highly efficient operation, while providing more performance headroom for graphics-intensive applications.

Impressive increases in data transfer rates

With PCIe Gen3 finally offered in Intel's low-power core processors, data rates have increased to 32 gigabytes (16 gigabytes coming in and going out) at 8GHz clock frequency, which is twice that of PCIe Gen2 at 5GHz.

Compatibility with USB 3.1 Gen2 for low-power x86 is another

advantage that ups performance to a maximum of 10Gbit/s, allowing UHD video to be transmitted as an uncompressed signal over USB. This is definitely a welcomed feature, for example, for transmitting raw data directly from a camera out to a display.

Native Ethernet support that unleashes full bandwidth

Support for time synchronised networking with Intel time coordinated computing and real-time systems hypervisor are further advantages. Compared to the previous generation, there is also native support for 3x 2.5GbE Ethernet, which drastically increases LAN bandwidth.

But real-time support over IP

is only one side of the equation in industrial applications requiring a deterministic behaviour. The same IP based communication between embedded computers and I/Os, with reduced latency and minimised jitter, is also required in synchronous processes. With the support of Intel's TCC technology, these can now be orchestrated as well.

The transportation industry's use of real-time applications that require integration of a CAN bus, for example, will benefit from these features. Settings can be fine-tuned with the Intel TCC software toolkit.

Advanced security inside

Security is a must-have feature for IIoT edge devices. Although it can be integrated by OEMs through the use of virtualisation technologies, as with the RTS real-time hypervisor for example, a more optimal solution is to have it as part of the hardware. The new low-power processor based boards and modules from congatec—which are designed for use with Intel's new Atom, Celeron, and Pentium processors—come with a wide range of integrated features that help to address high security demands. These include Intel Boot Guard 2.1 for boot verification, Intel Platform Trust Technology (Intel PTT) for compatibility with TPM, and Intel Dynamic Application Loader (Intel DAL) for helping developers create uniform and trustworthy applications.

For encrypting and decrypting data, congatec's Elkhart Lake based boards and modules provide even more value with the inclusion of Intel Advanced Encryption Standard New Instructions (AES-NI) and SHA Extensions that provide advanced security for secure hash algorithms that are hardware accelerated. SMx cipher support and HDCP 2.3 copy protection for the playback of HD media are additional features that will be appreciated by developers. **EFY**